

Application for Automated Bill of Quantities Generation & Extraction of Architectural Elements from 2D Floor Plans

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Abstract. The manual extraction of architectural elements from floor plans to generate a Bill of Quantities is a time consuming and error prone process within the construction industry. This project presents a desktop application designed to automate this workflow. Built using Python and the PySide6 framework, the software integrates deep learning semantic segmentation, Optical Character Recognition, and Large Language Models. A LangGraph based agent maps extracted elements to industry standard frameworks, estimating missing costs via a local database. An interactive chat assistant powered by a ReAct tool calling agent allows users to query floor plan metrics dynamically. The application was evaluated through a user survey where industry professionals reviewed a demonstration video of the software, yielding highly positive feedback regarding its utility and export capabilities. Additionally, an experimental attempt was made to improve the baseline CubiCasa5k segmentation model using Asymmetric Convolutions and a dual Attention Mechanism. While the application pipeline succeeded, the modified computer vision architecture yielded degraded segmentation results, highlighting the strict challenges of altering pre trained models.

Key words: PySide6, Quantity Takeoff, Bill of Quantities (BOQ), Large Language Models (LLM), CubiCasa5K, Architecture, Floor Plan Analysis, Semantic Segmentation

1. Introduction

The construction industry is historically slow to adopt automated digital workflows. A foundational element of any construction project is the Bill of Quantities. This document details the exact materials, parts, and finishes required to complete a build. To produce this document, quantity surveyors perform a Quantity Takeoff, which involves manually measuring dimensions, areas, and perimeters from architectural drawings. This process is highly repetitive, heavily reliant on human concentration, and vulnerable to costly calculation errors. A minor miscalculation in the area of a floor plan can compound into severe budget discrepancies during the procurement phase.

While Building Information Modelling offers three dimensional data extraction, the vast majority of initial tenders and smaller scale projects still rely on two dimensional raster images, PDF files, or CAD files for floor plans (Figure 3). Consequently, there is a significant industry need for tools capable of bridging the gap between flat two dimensional drawings and structured, costed data.

This project addresses these challenges by developing an end to end software application that automates the generation of a Bill of Quantities directly from two dimensional floor plan images. The overarching objective is to provide a comprehensive desktop tool that combines

computer vision for spatial recognition with Large Language Models for semantic reasoning and dynamic user interaction.

The goal was not merely to train a neural network in isolation, but to build a human in the loop system that quantity surveyors can integrate into their existing workflows. This involves managing material databases, editing artificial intelligence generated polygons, querying specific room data via a conversational interface, and exporting the final data into industry standard formats like Excel and Computer Aided Design (.dwg) files. To validate the success of this objective, the completed application was presented to twenty-four industry professionals via a demonstration video, and their feedback was collected through a detailed user research survey. This report documents the software engineering pipeline, the integration of advanced generative artificial intelligence agents, the evaluation feedback, and a critical analysis of the computer vision experiments conducted during the project.

2. Related Work

The automation of floor plan analysis and quantity surveying touches upon multiple domains, including computer vision, document parsing, and natural language processing.

2.1. Floor Plan Analysis and Semantic Segmentation

Early approaches to floor plan analysis relied on traditional computer vision heuristics, such as the Hough transform, to extract straight lines and detect corners based on hard-coded geometric rules [1]. However, these methods lacked robustness and struggled to generalise across the diverse drawing styles, noise, and artefacts found in real world architectural blueprints. Overcoming these limitations required a paradigm shift towards deep learning.

The foundation for modern dense pixel prediction was established by Long et al. [2] with Fully Convolutional Networks (FCN). This breakthrough was quickly adapted to architectural document analysis; for instance, Dodge et al. [3] successfully applied FCNs to segment walls and detect architectural objects across varying artistic styles. General semantic segmentation was subsequently improved by the U-Net architecture by Ronneberger et al. [4], which introduced skip connections to recover fine spatial details.

Building upon these foundational architectures, Kalervo et al. [5] introduced the CubiCasa5k dataset and a multi task convolutional neural network specifically tailored for floor plan analysis. Rather than a standard FCN or U-Net, their work utilises a Stacked Hourglass network, originally developed by Newell et al. [6] for human pose estimation. This architecture effectively captures complex spatial relationships across multiple scales through repeated downsampling and upsampling steps. Other notable modern works in this domain include DeepFloorplan by Zeng et al. [7] and the Raster to Vector transformation framework by Liu et al. [8].

2.2. Architectural Modifications and Loss Functions

To improve feature representation, researchers have explored various attention mechanisms. The Convolutional Block Attention Module infers attention maps sequentially along channel and spatial dimensions [9]. Squeeze and Excitation Networks model channel wise relationships to enhance representational power [10]. Modifying convolutional kernels has also proven beneficial. Ding et al. [11] proposed Asymmetric Convolution Blocks, which combine standard square convolutions with horizontal and vertical convolutions to strengthen the network against rotational distortions.

Handling class imbalance is a critical challenge in semantic segmentation. Lin et al. [12] introduced Focal Loss to address extreme foreground to background class imbalances by dynamically scaling the cross entropy loss based on confidence. Furthermore, Kendall et al. [13] proposed a framework for multi task learning using homoscedastic uncertainty, allowing networks to automatically weight different loss functions during training.

2.3. Optical Character Recognition and Generative AI

In parallel to computer vision, Optical Character Recognition has been widely applied to digitise technical documents. Systems like EasyOCR utilise deep learning to detect text in unconstrained environments [14], which is vital for reading room labels on blueprints.

Recently, Large Language Models (LLMs) have transformed data categorisation tasks. In the construction domain, researchers are exploring how these models can interpret complex project specifications [15]. Frameworks like LangChain have made it possible to deploy these models as reactive agents capable of executing external tools [16]. The ReAct paradigm allows language models to dynamically trigger Python functions to solve mathematical problems, which overcomes their inherent limitations in arithmetic calculations [17, 18].

3. Project Background and Context

A Bill of Materials lists all the materials, parts, and labour required for a project. Creating this document requires adherence to industry standards. Because regional and project specific requirements vary, quantity surveyors utilise different measurement frameworks, such as the Principles of Measurement International (POMI) [19], the Standard Method of Measurement (SMM7) [20], or the New Rules of Measurement (NRM2) [21]. These frameworks categorise work into specific hierarchical sections. For example, masonry might fall under Section D in POMI, whereas surface finishes or sanitary appliances might fall under Section N in SMM7.

Currently, professionals use software like CostX, Bluebeam Revu, or PlanSwift to measure floor plans. However, these tools still require the user to manually click the corners of every room to draw polygons. Standardising unstructured visual data from a floor plan into rigid categories automatically requires both geometric accuracy from a computer vision model and contextual understanding from a language model. The challenge lies in translating raw pixel coordinates and bounding boxes into a hierarchical dataset where a room entity contains specific flooring materials, wall finishes, and nested fixture elements.

4. Methodology and System Specification

The system architecture follows a linear pipeline divided into visual extraction, user interaction, data processing, and report generation as seen in the flowchart (Figure 1). The graphical user interface is built with PySide6 to provide a responsive and modern desktop experience following Microsoft’s fluent UI [22].

First, the user opens a floor plan as either an image or PDF. When the user clicks ‘Analyse’ the image is processed by a PyTorch based segmentation model to identify room boundaries, walls, and fixtures. An Optical Character Recognition module scans the image in four rotational directions to capture text labels accurately. The extracted polygons and text are merged and displayed on an interactive canvas.

4.1. Database and User Interaction

A local SQLite database stores material costs, brand names, and visual textures across distinct architectural categories. The user interface includes a custom database editor that supports direct clipboard image pasting using a custom item delegate. The main viewer allows users to calibrate the pixel to meter ratio by selecting two points and entering a known real world length. This calibration ratio is then injected into the global application state. Because the segmentation is not always perfect, the application provides a strong suite of manual override tools. Users can enter measurement mode to draw new polygons over missed rooms or fixtures. The system computes the area of these custom shapes and integrates them directly into the segmentation JSON payload, ensuring that manual corrections are treated identically to automated predictions during the report generation phase.

4.2. Generative AI Chat Assistant

A highly interactive chat panel was implemented in the application to provide users with dynamic data querying capabilities. This system leverages the LangChain framework to create a ReAct agent. The agent is provided with four specific tools: a materials database query tool, a unit conversion tool, a perimeter calculation tool, and a strict mathematical evaluation tool. Large Language Models are notoriously poor at performing arithmetic. Therefore, the system prompt explicitly forces the agent to use the calculation tool, which evaluates mathematical expressions securely in Python, for any numerical operation. The perimeter calculation tool is dynamically instantiated so that it inherits the user defined pixel to unit ratio from the active canvas.

A safety mechanism was engineered to prevent the language model from leaking raw JSON data to the user interface. If the agent attempts to output JSON, an interceptor loop wipes the agent memory, appends a strict corrective system prompt, and forces the model to regenerate a natural language response. This ensures a professional, conversational experience for the user.

4.3. Automated Report Generation

The reporting module operates using LangGraph to manage a multi step state graph workflow. To accommodate diverse regional requirements, the graphical export panel features a selection interface allowing the user to specify their desired measurement standard (such as POMI, SMM7, or NRM2) prior to generating the report. This selection is injected directly into the global report state.

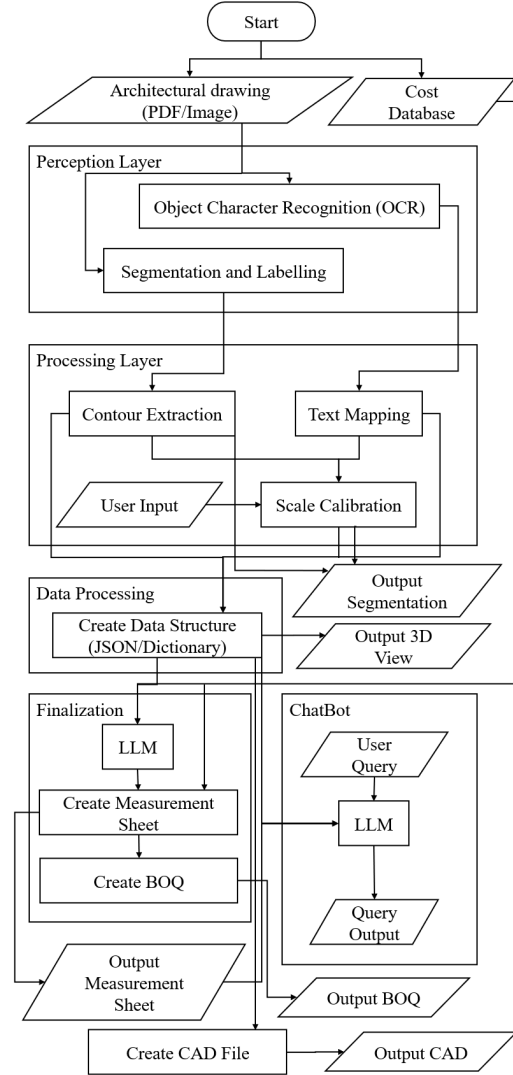


Figure 1: Application Flowchart

The workflow consists of four distinct nodes. First, the categorisation node aggregates the visual polygons and dynamically maps them to the specific sections of the user selected standard. If a material is unrecognised by the static mapping rules, the language model categorises it dynamically based on the active framework guidelines. Second, the unassigned items node invokes a ReAct agent to search the SQLite database. It extracts the mean, median, and mode prices for similar items and calculates a realistic estimated cost for any item missing a price. This is crucial for providing a complete baseline estimate even when the user has not explicitly assigned a material to every single room. Third, the preliminaries node generates standard preliminary costs, such as site setup and general requirements, tailored to the selected measurement framework. Finally, the compiler node structures this data into a comprehensive JSON format, which the application converts into stylised Excel and CSV files using the Pandas library.

Algorithm 1 ReAct Agent Workflow with Dynamic BOQ Standard Categorisation

Require: List of *UnassignedItems*, SQLite Database *DB*, Selected Standard *BOQ_{std}*

Ensure: List of *EstimatedItems* with calculated total costs

```

1: EstimatedItems  $\leftarrow \emptyset$ 
2: for each item in UnassignedItems do
3:   Category  $\leftarrow$  Map to Industry Section(item.name, BOQstd)
4:   AgentPrompt  $\leftarrow$  Construct ReAct prompt(item, Category, BOQstd)
5:   while Agent needs more information do
6:     Query  $\leftarrow$  Agent(Action: query_materials_database)(Category)
7:     Results  $\leftarrow$  DB.execute(Query)
8:     Agent  $\leftarrow$  Update Context(Results)
9:   end while
10:  Rate  $\leftarrow$  Agent(Calculate mode and median from Results)
11:  Quantity  $\leftarrow$  item.quantity
12:  TotalCost  $\leftarrow$  Agent(Action: calculate_item_total)(Quantity, Rate)
13:  item.rate  $\leftarrow$  Rate
14:  item.total  $\leftarrow$  TotalCost
15:  EstimatedItems  $\leftarrow$  EstimatedItems  $\cup$  {item}
16: end for
17: return EstimatedItems

```

4.4. Security and Privacy

The overall architecture of the application prioritises data security and user confidentiality by executing the majority of its computational pipeline locally. The PyTorch semantic segmentation model, the Optical Character Recognition module, and the SQLite database management operate entirely on the host machine. This architecture ensures that sensitive architectural floor plans and proprietary material pricing data never leave the user’s local environment. Furthermore, the deep learning models were trained using the CubiCasa5k dataset. Because this is a publicly available and open source repository of floor plans, it mitigates any ethical or legal concerns regarding the use of confidential, real world architectural drawings during the model training phase.

The only component of the application capable of requesting an external internet connection is the generative artificial intelligence chat assistant and automated reporting agent. To provide users with absolute control over their data privacy, the application implements a dynamic model selection framework. Users who require advanced reasoning can securely

input their own API keys to connect with external cloud providers such as OpenAI, Google, Anthropic, or OpenRouter. However, to maintain a strict zero trust security posture, the system defaults to local inference. By integrating seamlessly with Ollama, the application allows users to run Large Language Models entirely offline. This ensures that even dynamic conversational queries, data categorisation, and automated cost estimations can be performed without ever transmitting project specifications to third party cloud servers.

5. Experiments and Evaluation

An evaluation phase was conducted with practicing quantity surveyors and estimators (Figure 2a), mostly in the residential and commercial construction sector (Figure 2b). This phase served as a critical review of the nearly finished application to assess its utility in a real world context.

5.1. User Research

A comprehensive demonstration video was produced, showcasing the end to end capabilities of the application (see Appendix B). The video highlighted the floor plan ingestion, the automated segmentation, the manual measurement overrides, the LangChain chat assistant, and the final generation of the Bill of Quantities in Excel and CAD formats. This video was distributed to twenty-four industry professionals alongside a structured survey designed to gather both quantitative ratings and qualitative feedback (see Appendix C). Furthermore, a guided usability session was conducted with a Commercial Manager and a Senior Quantity Surveyor to evaluate the application’s practical utility.

5.1.1 Survey Results and Industry Feedback

The feedback gathered from the survey was overwhelmingly positive and validated several key engineering decisions made during the project.

Respondents reacted highly favourably to the automated Quantity Takeoff generation (Figure 7c). Many noted that the ability to export the extracted data seamlessly into Microsoft Excel and DXF files aligned perfectly with their current software ecosystems (Figure 3c). The integration of the different standards directly into the export logic was highlighted as a nice to have time saving feature (Figure 5).

The generative AI Chatbot received strong praise. Professionals appreciated the ability to query material costs and ask for specific calculations, such as the total perimeter of all bedrooms, without needing to manually filter through the exported spreadsheets (Figure 7b). The implementation of the Python calculation tools within the LangChain agent proved effective, as users noted the accuracy of calculations to be an essential attribute (Figure 4).

Crucially, the survey feedback validated the inclusion of the manual measurement tools. Industry professionals explicitly stated that being able to handle complex or degraded architectural drawings is highly desired and that AI chat queries aren’t critical but is highly valuable (Figure 5, Figure 7b). The usability session confirmed that the ability to manually draw missing polygons, correct labels, and update the database within the same application window was an essential requirement for adoption. The seamless merging of human drawn polygons with the AI generated data was considered a major success of the system design.

5.2. Experiments and Computer Vision Modifications

While the software pipeline successfully handled the data and received positive user feedback, the core of the extraction relies heavily on the accuracy of the underlying computer vision model. Experiments were conducted to improve the baseline CubiCasa5k segmentation model. The intention was to enhance the feature extraction capabilities of the network and address the severe class imbalance inherent in floor plan images as several small items such as fire places, bathtubs, and chimneys were noted to have very low Intersection Over Union (IoU) scores and accuracy. Due to hardware limitations, the training and evaluation of these models were offloaded to the University of Birmingham BlueBEAR High Performance Computing cluster.

5.2.1 Architectural Changes

Several architectural changes were made to the baseline Hourglass model in the PyTorch backend. First, Asymmetric Convolutions were introduced. The standard three by three convolutions within the residual blocks were replaced with a custom module. This module computes a standard three by three, a one by three horizontal filter, and a three by one vertical filter simultaneously. The outputs are summed together. The hypothesis was that explicit horizontal and vertical kernels would better capture the orthogonal nature of walls, doors, and windows in floor plans.

Second, an Attention Mechanism was inserted at the bottleneck of the network. This module concatenates a Channel Attention Module (CAM) and a Spatial Attention Module (SAM). The Channel Attention Module uses both adaptive average pooling and adaptive max pooling to squeeze the spatial dimensions, followed by a shared multi layer perceptron to infer channel wise dependencies. The Spatial Attention Module applies convolutions with varying dilation rates to capture multi-scale spatial context. The attention context is added to the feature map using a residual connection, ensuring that the original pre-trained features are preserved while being augmented by the attention weights.

5.2.2 Custom Loss Function

The loss function was fundamentally rewritten to address the specific challenges of floor plan segmentation. The baseline model uses standard cross entropy loss combined with multitask homoscedastic uncertainty. To combat the severe class imbalance where background pixels dominate the image, Focal Loss was implemented for the room and icon prediction heads. The focal loss dynamically scales the cross entropy loss by a factor of $(1 - p)^\gamma$, where p is the model confidence. This forces the model to focus on misclassified, difficult examples. The focal loss was integrated with the existing homoscedastic uncertainty parameters, creating a novel loss function that attempts to balance task uncertainty with class imbalance weighting.

5.2.3 Transfer Learning and Weight Surgery

To prevent overfitting on the limited dataset, pre-trained weights from the original CubiCasa5k model were loaded. Because the architecture had been modified, standard weight loading was impossible. A custom weight surgery script was implemented to copy the old square kernel weights into the main branch of the new asymmetric convolutions, initialising the new horizontal and vertical kernels to zero. A freezing strategy was employed where all pre-trained layers were frozen for a warmup period of ten epochs, allowing only the new attention layers to learn. Following the warmup, all layers were unfrozen, and the learning rate was reduced by a factor of ten for fine-tuning.

Algorithm 2 Multi-Task Focal Loss with Homoscedastic Uncertainty

Require: Predictions $\hat{y}$, Ground Truth y , Focal parameters γ, α

Require: Learnable uncertainty variances $\sigma_1^2, \sigma_2^2, \dots, \sigma_n^2$

Ensure: Total Loss $\mathcal{L}_{total}$

```
1: Function FocalLoss( $\hat{y}_c, y_c$ )
2:  $CE \leftarrow -\log(\hat{y}_c)$  ▷ Cross Entropy for target class
3:  $p_t \leftarrow \exp(-CE)$  ▷ Model confidence
4:  $\mathcal{L}_{focal} \leftarrow \alpha \cdot (1 - p_t)^\gamma \cdot CE$ 
5: return  $\mathcal{L}_{focal}$ 
6: End Function
7:  $\mathcal{L}_{rooms} \leftarrow \text{FocalLoss}(\hat{y}_{rooms}, y_{rooms})$ 
8:  $\mathcal{L}_{icons} \leftarrow \text{FocalLoss}(\hat{y}_{icons}, y_{icons})$ 
9:  $\mathcal{L}_{heatmaps} \leftarrow \text{MeanSquaredError}(\hat{y}_{heat}, y_{heat})$ 
10:  $\mathcal{L}_{total} \leftarrow 0$ 
11:  $Losses \leftarrow [\mathcal{L}_{rooms}, \mathcal{L}_{icons}, \mathcal{L}_{heatmaps}]$ 
12: for  $i \leftarrow 1$  to  $\text{Length}(Losses)$  do
13:    $\text{Weight} \leftarrow \frac{1}{2 \exp(\log(\sigma_i^2))}$ 
14:    $\text{Regularisation} \leftarrow \frac{1}{2} \log(\sigma_i^2)$ 
15:    $\mathcal{L}_{total} \leftarrow \mathcal{L}_{total} + (\text{Weight} \cdot Losses[i] + \text{Regularisation})$ 
16: end for
17: return  $\mathcal{L}_{total}$ 
```

6. Results

6.1. Software Pipeline Evaluation

As evidenced by the user survey, the software engineering aspect of the project was highly successful. The integration of the PySide6 frontend, the LangChain agent, and the SQLite database created a robust and genuinely useful tool. The application directly addresses the pain points identified by quantity surveyors. The seamless transition from an uploaded image to a fully costed, standard compliant Excel spreadsheet demonstrates a successful application of generative artificial intelligence in a traditional industry, such that most participants said they are likely to adopt the application into their daily workflow in its current state, as seen in Figure 6.

6.2. Computer Vision Experiment Results

Despite the theoretical soundness of the architectural modifications, the experimental computer vision model yielded degraded segmentation results compared to the baseline. When evaluated on the validation set, the modified architecture containing the Asymmetric Convolutions and the Attention Modules performed noticeably worse. The training loss plateaued prematurely, and the validation loss indicated a drop in overall pixel accuracy.

Several technical hypotheses explain this failure. First, introducing new, randomly initialised parameters at the most sensitive part of the network (the bottleneck) may have disrupted the spatial embeddings. Even with the residual connections and the warmup period, the attention mechanism likely forced the decoder to interpret the features differently, destroying the delicate representations learned during pre-training.

Second, combining Focal Loss with Homoscedastic Uncertainty likely caused severe gradient instability. The homoscedastic uncertainty parameter attempts to minimise the overall loss by scaling down the weight of difficult tasks. At the same time, the Focal Loss

attempts to amplify the loss of difficult pixels. These conflicting mathematical forces likely caused the network to exploit the uncertainty variance parameter to scale down the focal loss artificially, leading to a vanishing gradient problem where the model stopped learning entirely.

Finally, the CubiCasa5k dataset may not be large enough to train the additional parameters introduced by the Asymmetric Convolutions without aggressive regularisation. The model likely overfitted to the training set early in the fine-tuning phase.

Despite the overall shortcomings of the new architecture, the modifications did achieve their main goal. The aim was to improve the Intersection over Union and accuracy for smaller objects that have less training data. The experimental results show a slight to moderate increase in scores for these specific items. For example, the model performed better on the fire place, bathtub, chimney, door, toilet, and sauna bench. However, this localised improvement came at a clear cost. The gains for these smaller points caused a direct drop in the accuracy and Intersection over Union scores for most of the other common rooms and items.

Table 1: Overall Room and Item Metrics Comparison

Task	CubiCasa5k				Modified			
	Over. Acc	Mean Acc	Mean IoU	FreqW Acc	Over. Acc	Mean Acc	Mean IoU	FreqW Acc
Room seg	82.2	69.3	56.7	71.4	79.8	64.2	52.5	67.9
Icon seg	97.4	55.2	50.9	95.1	97.2	57.5	51.8	94.7

Table 2: Specific Room Metrics

Room Type	CubiCasa5k		Modified	
	IoU	Acc	IoU	Acc
Background	87.1	93.1	84.2	93.5
Outdoor	63.3	77.3	60.5	71.4
Wall	70.8	81.6	70.3	83.0
Kitchen	65.1	81.1	61.4	78.7
Living Room	65.7	82.0	62.2	77.0
Bedroom	73.9	84.3	68.7	81.7
Bath	56.6	65.8	56.5	72.7
Hallway	55.9	72.2	51.9	66.6
Railing	20.6	24.3	20.5	24.1
Storage	45.2	57.6	36.8	45.0
Garage	33.5	52.4	21.5	25.6
Other rooms	42.2	60.3	35.4	51.4

Table 3: Specific Item Metrics

Item Type	CubiCasa5k		Modified	
	IoU	Acc	IoU	Acc
Empty	97.4	99.4	97.1	99.3
Window	64.4	69.8	58.5	64.5
Door	49.5	54.1	51.2	57.5
Closet	67.6	78.3	65.4	73.6
Electr. Appl.	62.6	69.3	58.4	67.2
Toilet	58.3	62.0	61.5	68.2
Sink	52.0	59.7	49.2	58.5
Sauna bench	62.0	65.2	62.8	68.7
Fire Place	25.3	26.9	33.7	38.3
Bathtub	15.6	17.0	20.6	24.0
Chimney	5.1	5.2	11.8	12.8

7. Conclusion

This project successfully demonstrates the feasibility of an automated Bill of Quantities generation system. By combining semantic segmentation, Optical Character Recognition, and generative AI agents within a robust PySide6 interface, the application offers a practical solution to a major bottleneck in quantity surveying. The post development user research survey verified the success of the software, with industry professionals confirming the utility of the export features, the necessity of the manual override tools, and the efficiency of the conversational chat assistant.

The experimental computer vision phase highlighted the significant challenges associated with modifying established neural network architectures. While theories regarding Asymmetric Convolutions and Focal Loss are sound in isolation, integrating them into a multi task framework with pre trained weights requires careful mathematical balancing. Ultimately, the successful delivery of the software pipeline, combined with the clear roadmap for future feature expansions, establishes a strong foundation for integrating artificial intelligence into traditional estimation workflows.

8. Future Work

While the current iteration of the software successfully automates the foundational Quantity Takeoff process, the user research survey highlighted several feature suggestions that fall outside the scope of this project (Figure 8). These suggestions form a robust roadmap for future development:

8.1. External Databank Integration

Currently, the application relies on a local SQLite database for material pricing. However, industry standard cost data (such as BCIS or Spon's) is highly proprietary. Survey respondents, expressed a desire for the software to interface with existing Enterprise Resource Planning systems. Future versions should implement secure RESTful API with OAuth authentication to fetch real time, proprietary market rates from external subscription services. This would allow the application to bypass the local SQLite database entirely and generate the Bill of Quantities using a company's centrally managed, live ERP rate library.

8.2. Automated Validation and Cross-Checking

Future iterations could introduce an automated validation layer. This feature would cross-check already existing reports against the project data or standard architectural ratios to flag potential anomalies.

8.3. Revision Tracking and Delta Estimation

A requested feature from an industry professional was the ability to handle design revisions. Future work will explore implementing computer vision image diffing algorithms or structural similarity indices to compare an original design drawing with a revised drawing automatically. By isolating the geometric changes, the generative artificial intelligence workflow could be adapted to prepare a specific 'Variations' Bill of Quantities, pricing only the delta between the two design iterations.

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Declaration

AI tools such as Google Gemini and GitHub Copilot were used for idea generation, researching, and drafting code snippets. All content was reviewed, verified, and edited.

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Appendix

A. Project GitLab Repository

The source code for this project is available on GitLab [here](#).

B. Demonstration Video

The demonstration video of the application which was shown to the participants of the user research survey can be seen [here](#). Note that this is a slightly older version of the application.

C. Survey Responses

The survey questions can be found [here](#), and the raw survey responses can be found [here](#).



Figure 2: About participants of user research survey

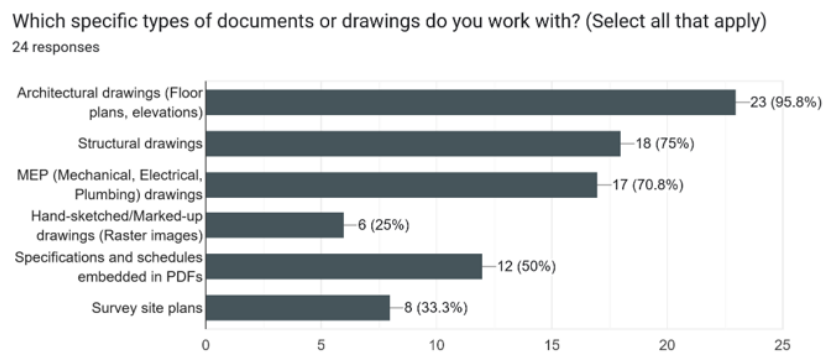
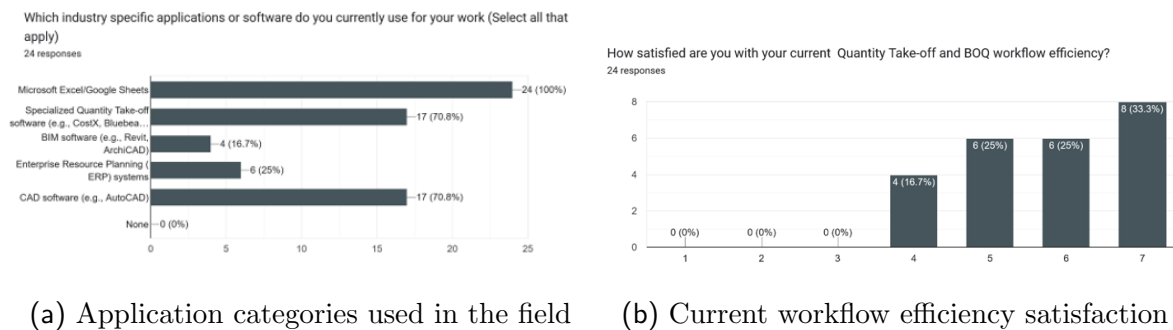


Figure 3: Current setup and workflow

Please rate the importance of the following general features in the application shown in the video:

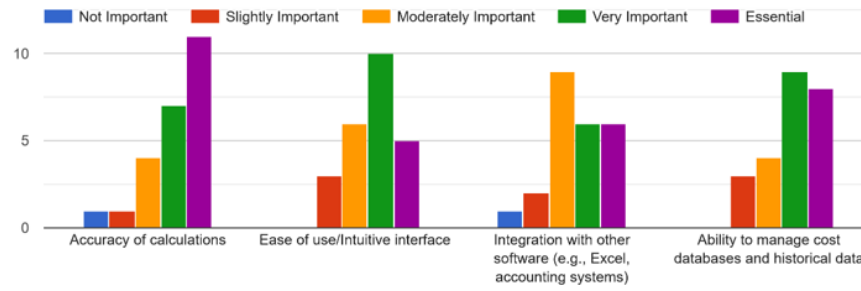


Figure 4: Importance of general features/attributes

If this application is available to you, what specific features would be most critical for you?

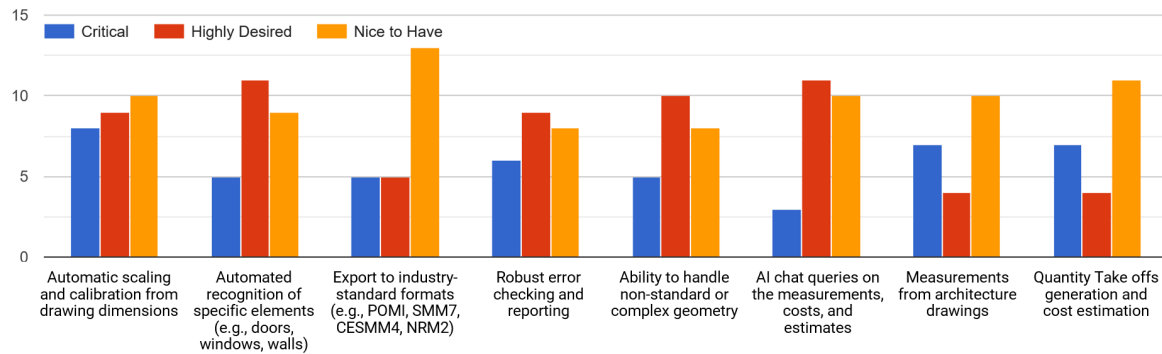


Figure 5: Desirability of specific features

If this application is available to you, how likely are you to adopt this application into your daily workflow in its current state?

24 responses

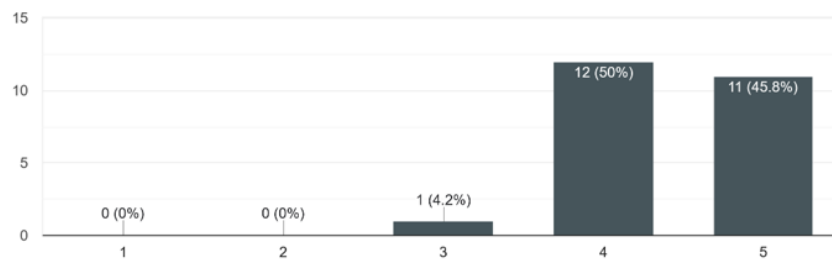
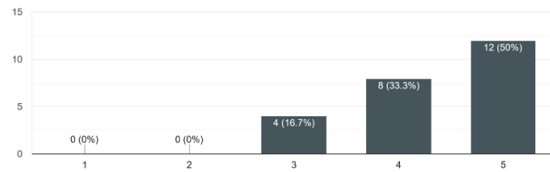


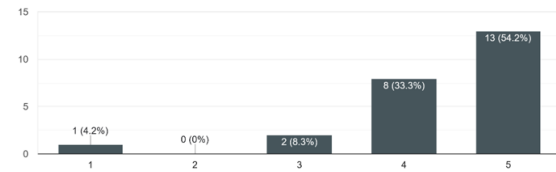
Figure 6: Likelihood of adoption of the application in its current state

Considering the nature of manual data entry, how helpful is the idea of a tool that automatically reads and extracts room text (OCR) for you?
24 responses



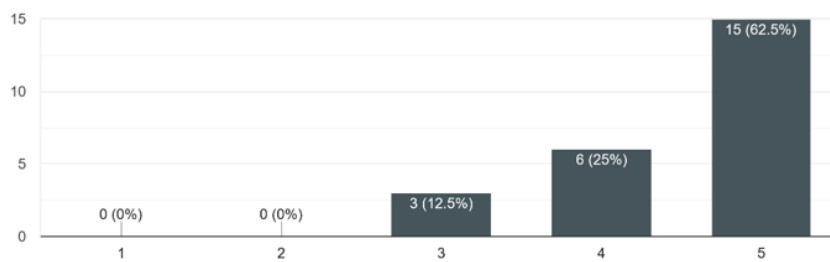
(a) Automatic OCR Text Extraction

How valuable is the idea of having an intelligent AI chatbot built directly into the software, allowing you to instantly ask questions like "What is the to...?" instead of manually adding up spreadsheet rows?
24 responses



(b) AI Chatbot to ask questions

Consider the feature to automatically generate Quantity Take-off and BOQs drafts directly from raster images (scans) and PDF drawings. How valuable to your daily work as a starting template?
24 responses



(c) Using generated QTO & BOQ as starting Template

Figure 7: Valability of specific features

Please share your thoughts on the application. How can it be improved?

10 responses

Integrate AI into the application with history data for benchmarking cost purposes.

It can be improved by providing more features

An option to link an existing available rate databank to the drawing or raster.

Any feature to include validation and cross checking of generated reports

Very good application

It is helpful and easy to use

This application is very useful for preparing estimation and costing efficiently.

Add the feature of comparing an original design drawing with the revised drawing to identify the changes and prepare BOQ of the changes.

This application is okay

pricing feature would be a nice additional feature

Figure 8: User thoughts, feedback, and feature requests